中国科学院计算技术研究所
INSTITUTE OF COMPUTING TECHNOLOGY, CHINESE ACADEMY OF SCIENCES

A3: 384-Node Supercomputing System

Huawei CloudMatrix384 SuperNode

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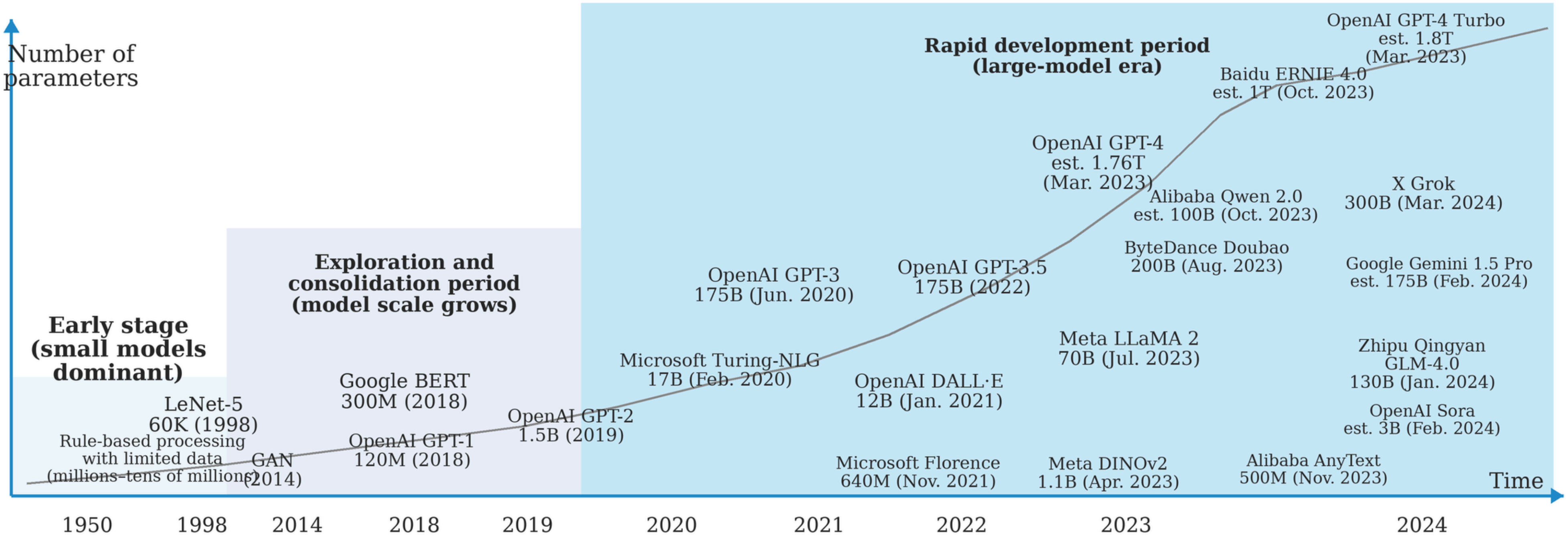
Huawei Tau (τ) Scaling Law

Introduction

Model Parameter



Overview

CloudMatrix384
A Supernode for AI
遥遥领先
遥遥领先，未来已来！
1
384 AI processors work together
2
Ultra-fast links help them talk quickly
3
Acts like one giant computer team
4
Built for training and running big AI models
Think of it like a huge team project:
many smart workers
one fast network
one big goal
Big, fast, and designed for the AI era.

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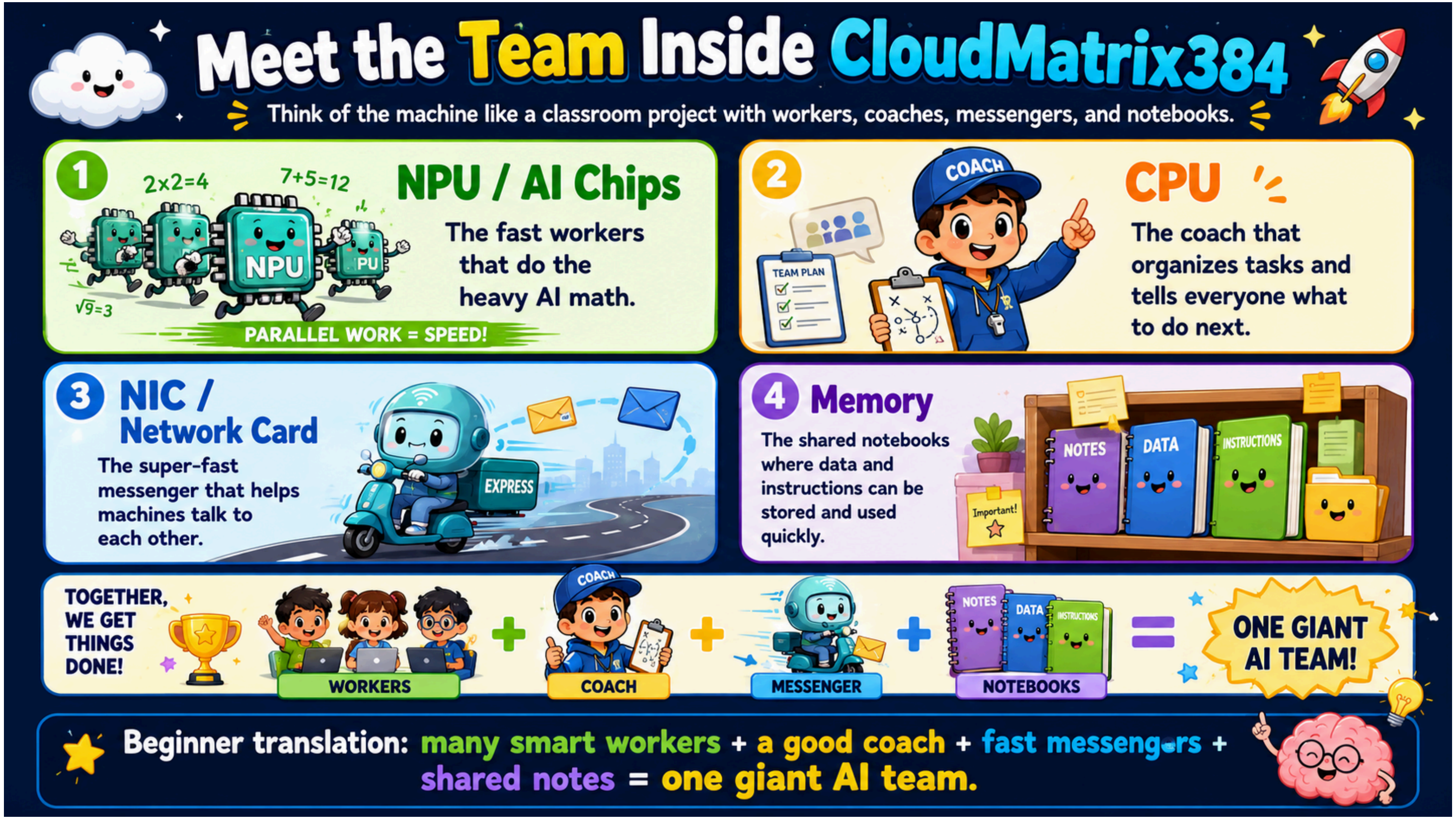
> Key Idea: AI speed is not only about a powerful chip. It is also about how well many chips cooperate.

For Example:

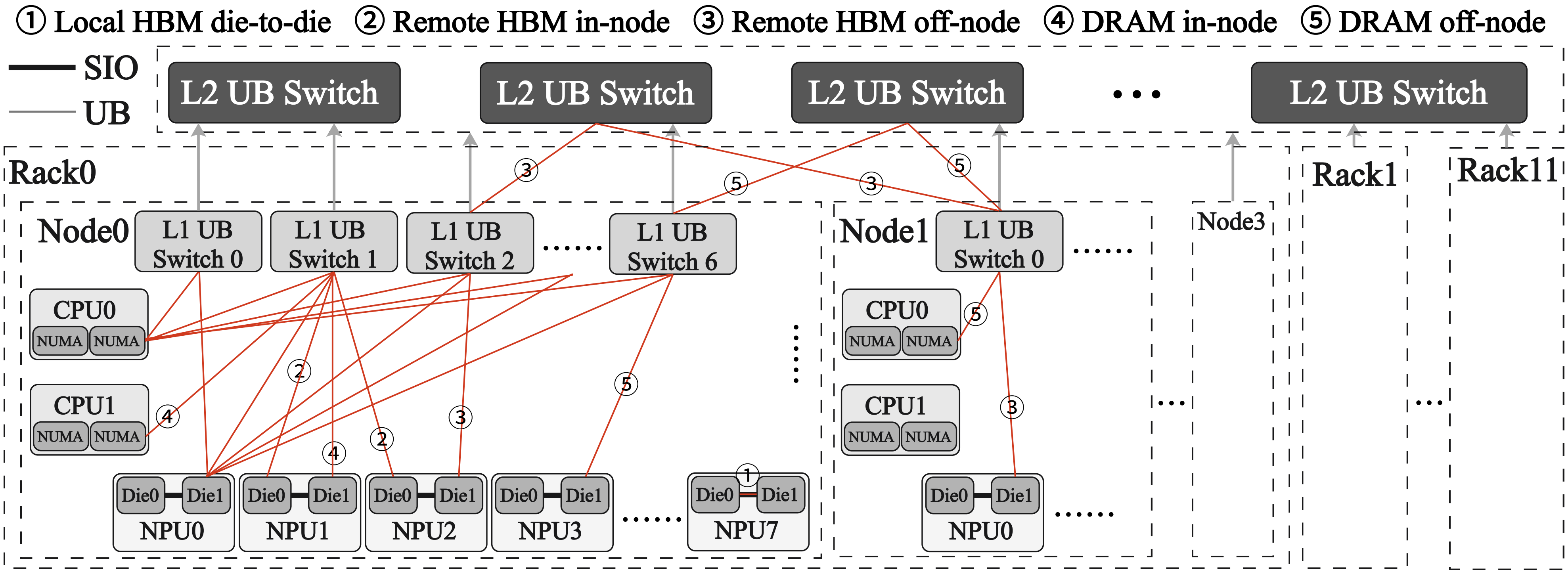
- One student = slow for a giant task
- 384 AI workers + fast chatting = one super team
- CloudMatrix384 is the machine version of this teamwork

> Research Question: **How to scale the AI performance by cooperation Hardware and Software?**

Introduction

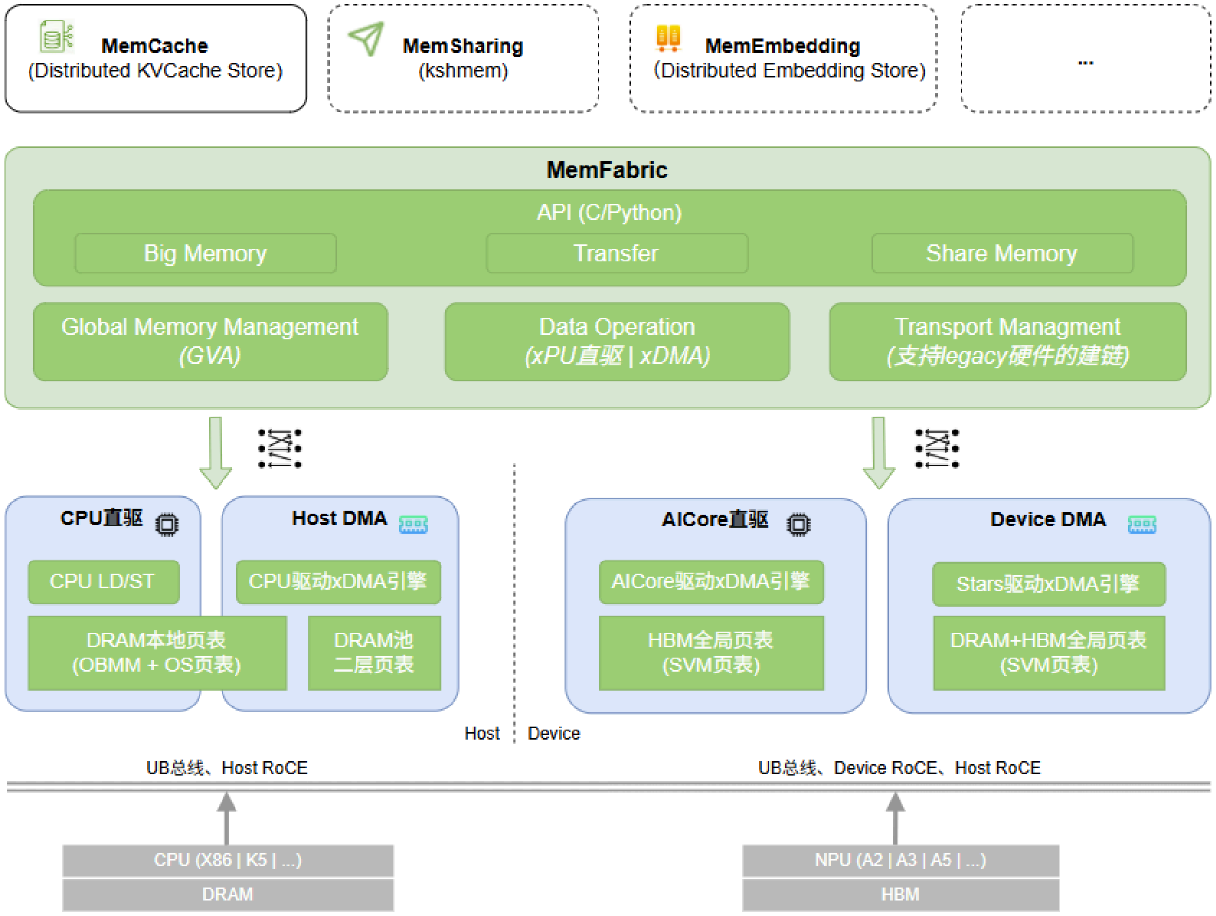


Huawei CloudMatrix384



Design

Overview



MemCache
(Distributed KVCache Store)
MemSharing
(kshmem)
MemEmbedding
(Distributed Embedding Store)
...
MemFabric
API (C/Python)
Big Memory
Transfer
Share Memory
Global Memory Management
(GVA)
Data Operation
(xPU直驱 | xDMA)
Transport Managment
(支持legacy硬件的建链)
CPU直驱
CPU LD/ST
Host DMA
CPU驱动xDMA引擎
DRAM本地页表
(OBMM + OS页表)
DRAM池
二层页表
AICore直驱
AICore驱动xDMA引擎
HBM全局页表
(SVM页表)
Device DMA
Stars驱动xDMA引擎
DRAM+HBM全局页表
(SVM页表)
Host
Device
UB总线、Host RoCE
UB总线、Device RoCE、Host RoCE
CPU (X86 | K5 | ...)
DRAM
NPU (A2 | A3 | A5 | ...)
HBM

Experiment

Table 2. Overall prefill throughput (per accelerator) for DeepSeek-R1.

Method	**Batch Size**	**Input Length**	**Throughput (tokens/s)**	**Throughput per TFLOPS**
DeepSeek on H800 (Blog)	N/A	N/A	4,026	2.03
SGLang on H100 (Default)	16,384	4,096	6,288	3.18
CloudMatrix-Infer (Default)	16,384	4,096	5,655	3.76
DeepSeek on H800 (Profile)	16,384	4,096	**7,839**	3.96
SGLang on H100 (Perfect EPLB)	16,384	4,096	7,417	3.75
CloudMatrix-Infer (Perfect EPLB)	16,384	4,096	6,688	**4.45**

Experiment

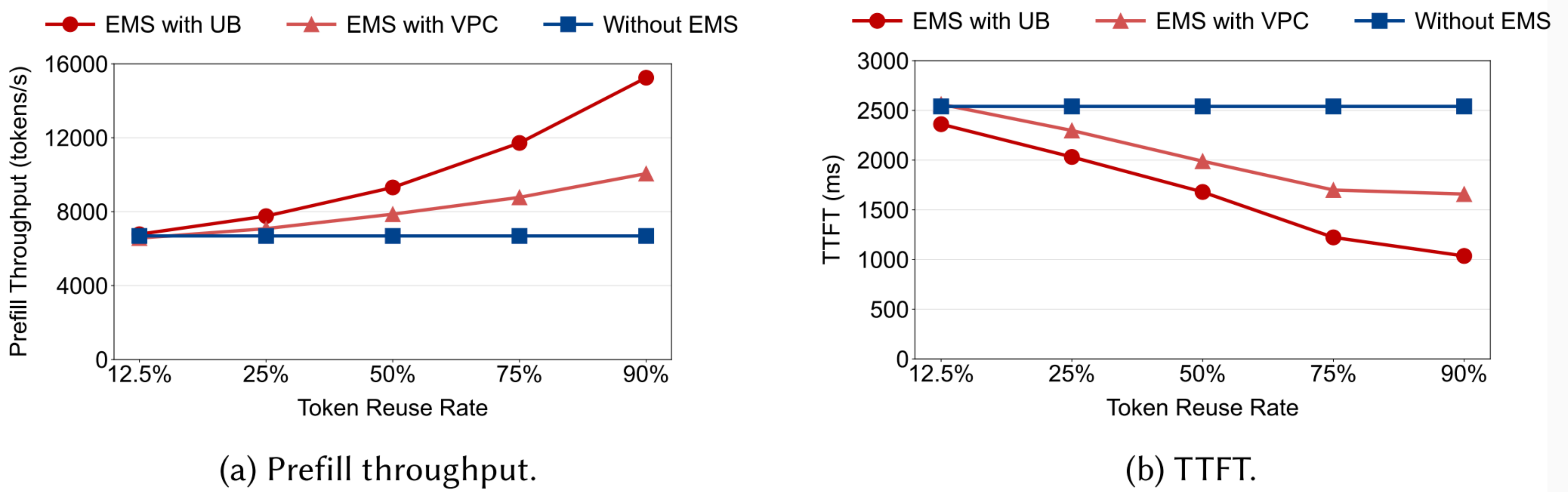


Fig. 23. The overall prefill throughput and TTFT using EMS-Context Caching with different configurations.

Takeaways & Discussion

Conclusion

- Collectively, our findings validate Huawei CloudMatrix as a highly effective, scalable, and performance-optimized platform for deploying large-scale AI workloads, setting a benchmark for future AI datacenter infrastructures.

Takeaways & Discussion

- Why should non-CS students care?
  - New Competition
    - Countries and companies compete not only on chips, but also on system design and networking
  - Real Tradeoffs
    - More power also means more energy, cooling, cost, and engineering complexity

Takeaways & Discussion

- For CS students,
  - Hardware Trends:
    - CloudMatrix384 focuses on full interconnectivity and resource pooling, not just adding more NPUs.
  - Abstraction Trends:
    - MemFabric uses GVA + xcopy to transform remote HBM/DRAM into memory resources that can be directly expressed by applications.
  - System Opportunities:
    - Memory-intensive workloads such as KV Cache, Embedding, and parameter caching will benefit first.
  - Research Space:
    - Placement, prefetching, consistency, and runtime scheduling still require workload-aware design.

Thank you!

https://gitcode.com/Ascend/memfabric_hybrid

Q & A